

# Skill Week-12

## Session-23

**Task 1:** To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance.

```
lenovo@Manya-$ mkdir -p OSSP_Project
lenovo@Manya-$ cd OSSP_Project
lenovo@Manya-/OSSP_Project$ nano data.txt
lenovo@Manya-/OSSP_Project$ cat data.txt | grep Linux
Linux is an operating system
Linux supports processes
Linux supports pipes
Linux supports signals
Linux supports system calls
lenovo@Manya-/OSSP_Project$ cat data.txt | grep Linux | sort | uniq | wc -l
5
lenovo@Manya-/OSSP_Project$ cat data.txt | grep -i process | sort
Linux supports processes
Processes execute programs
Signals notify processes
lenovo@Manya-/OSSP_Project$ sleep 30 &
sleep 40 &
sleep 50 &
[1] 4141
[2] 4147
[3] 4153
lenovo@Manya-/OSSP_Project$ jobs
[1]  Running                  sleep 30 &
[2]-  Running                  sleep 40 &
[3]+  Running                  sleep 50 &
lenovo@Manya-/OSSP_Project$ ps
  PID TTY          TIME CMD
 3946 pts/0        00:00:00 bash
 4141 pts/0        00:00:00 sleep
 4147 pts/0        00:00:00 sleep
 4153 pts/0        00:00:00 sleep
 4176 pts/0        00:00:00 ps
lenovo@Manya-/OSSP_Project$ kill %1
[1]  Terminated              sleep 30
lenovo@Manya-/OSSP_Project$ jobs
[2]-  Running                  sleep 40 &
[3]+  Running                  sleep 50 &
lenovo@Manya-/OSSP_Project$ time ls -R /usr > /dev/null
ls: cannot open directory '/usr/lib/modules/6.18.33.2-microsoft-standard-WSL2/lost+found': Permission denied

real    0m7.859s
user    0m0.146s
sys     0m1.218s
[2]-  Done                    sleep 40
```

```

akshitha_reddy@Akshitha-Reddy:~/OSSP_Project$ top
top - 15:04:43 up 1:13, 1 user, load average: 0.11, 0.03, 0.01
Tasks: 31 total, 1 running, 30 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.
MiB Mem : 5769.6 total, 5028.1 free, 551.4 used, 337.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 5218.2 avail Mem

```

| PID  | USER     | PR | NI | VIRT    | RES   | SHR   | S | %CPU | %MEM | TIME+   |
|------|----------|----|----|---------|-------|-------|---|------|------|---------|
| 85   | root     | 20 | 0  | 35600   | 12544 | 9232  | S | 0.7  | 0.2  | 0:15.32 |
| 1    | root     | 20 | 0  | 24232   | 15444 | 11556 | S | 0.0  | 0.3  | 0:01.10 |
| 2    | root     | 20 | 0  | 3180    | 2208  | 2072  | S | 0.0  | 0.0  | 0:00.01 |
| 7    | root     | 20 | 0  | 3180    | 2048  | 1940  | S | 0.0  | 0.0  | 0:00.00 |
| 47   | root     | 19 | -1 | 50424   | 16924 | 15592 | S | 0.0  | 0.3  | 0:00.41 |
| 79   | systemd+ | 20 | 0  | 22416   | 14440 | 11944 | S | 0.0  | 0.2  | 0:00.11 |
| 153  | root     | 20 | 0  | 159348  | 1884  | 1556  | S | 0.0  | 0.0  | 0:00.02 |
| 154  | root     | 20 | 0  | 454144  | 13556 | 1516  | S | 0.0  | 0.2  | 0:02.85 |
| 159  | root     | 20 | 0  | 159348  | 1880  | 1556  | S | 0.0  | 0.0  | 0:00.00 |
| 187  | root     | 20 | 0  | 2888    | 2012  | 1884  | S | 0.0  | 0.0  | 0:00.05 |
| 188  | root     | 20 | 0  | 4472    | 3116  | 2872  | S | 0.0  | 0.1  | 0:00.02 |
| 189  | message+ | 20 | 0  | 8820    | 5376  | 4664  | S | 0.0  | 0.1  | 0:00.11 |
| 193  | root     | 20 | 0  | 44096   | 29688 | 14544 | S | 0.0  | 0.5  | 0:00.25 |
| 196  | root     | 20 | 0  | 2220492 | 39504 | 25344 | S | 0.0  | 0.7  | 0:03.34 |
| 197  | root     | 20 | 0  | 18440   | 9452  | 8204  | S | 0.0  | 0.2  | 0:00.13 |
| 214  | syslog   | 20 | 0  | 220548  | 5644  | 4504  | S | 0.0  | 0.1  | 0:00.10 |
| 230  | root     | 20 | 0  | 5492    | 2796  | 2612  | S | 0.0  | 0.0  | 0:00.01 |
| 244  | _chrony  | 20 | 0  | 20512   | 7784  | 6952  | S | 0.0  | 0.1  | 0:00.12 |
| 252  | root     | 20 | 0  | 123460  | 32648 | 18036 | S | 0.0  | 0.6  | 0:00.24 |
| 262  | _chrony  | 20 | 0  | 12096   | 2432  | 1788  | S | 0.0  | 0.0  | 0:00.00 |
| 393  | root     | 20 | 0  | 8644    | 5596  | 4756  | S | 0.0  | 0.1  | 0:00.03 |
| 437  | akshith+ | 20 | 0  | 22772   | 13272 | 10800 | S | 0.0  | 0.2  | 0:00.21 |
| 439  | akshith+ | 20 | 0  | 22920   | 4076  | 2076  | S | 0.0  | 0.1  | 0:00.00 |
| 467  | akshith+ | 20 | 0  | 6268    | 5548  | 3924  | S | 0.0  | 0.1  | 0:00.04 |
| 1927 | polkitd  | 20 | 0  | 306444  | 8504  | 7692  | S | 0.0  | 0.1  | 0:00.21 |
| 3942 | root     | 20 | 0  | 3188    | 1120  | 980   | S | 0.0  | 0.0  | 0:00.00 |
| 3943 | root     | 20 | 0  | 3204    | 1260  | 1108  | S | 0.0  | 0.0  | 0:00.02 |
| 3946 | akshith+ | 20 | 0  | 6784    | 6176  | 3936  | S | 0.0  | 0.1  | 0:00.10 |
| 4247 | akshith+ | 20 | 0  | 8172    | 6036  | 3932  | R | 0.0  | 0.1  | 0:00.00 |
| 4248 | root     | 20 | 0  | 35604   | 6616  | 3272  | S | 0.0  | 0.1  | 0:00.00 |
| 4249 | root     | 20 | 0  | 35604   | 6616  | 3272  | S | 0.0  | 0.1  | 0:00.00 |

```

[3]+  Done                  sleep 50

```

```

lenovo@Manya~/OSSP_Project$ free -h
               total        used        free      shared  buff/cache   available
Mem:           5.6Gi        548Mi        4.9Gi        3.8Mi        337Mi        5.1Gi
Swap:          2.0Gi           0B        2.0Gi

lenovo@Manya~/OSSP_Project$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            2.9G   0 2.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            2.9G  4.0K 2.9G   1% /mnt/wsl
drivers          307G  136G  172G  45% /usr/lib/wsl/drivers
/dev/sdd        1007G   2.7G  954G   1% /
none            2.9G   52K 2.9G   1% /mnt/wslg
none            2.9G   0 2.9G   0% /usr/lib/wsl/lib
rootfs          2.9G   2.8M 2.9G   1% /init
none            2.9G  800K 2.9G   1% /run
none            2.9G   0 2.9G   0% /run/lock
none            2.9G   0 2.9G   0% /run/shm
none            2.9G   80K 2.9G   1% /mnt/wslg/versions.txt
none            2.9G   80K 2.9G   1% /mnt/wslg/doc
C:\             307G  136G  172G  45% /mnt/c
D:\             169G  130G   40G  77% /mnt/d
tmpfs            2.9G   0 2.9G   0% /tmp
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0 1.0M   0% /run/credentials/systemd-resolved.service
snapfuse         4.5M  4.5M   0 100% /snap/tree/54
snapfuse         51M   51M   0 100% /snap/snapd/27591
snapfuse         74M   74M   0 100% /snap/core22/2411
none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs            577M  12K  577M   1% /run/user/1000

lenovo@Manya~/OSSP_Project$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          48 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 8
On-line CPU(s) list:    0-7
Vendor ID:              AuthenticAMD
Model name:             AMD Ryzen 5 7235HS
CPU family:             25
Model:                 68
Thread(s) per core:     2
Core(s) per socket:     4
Socket(s):              1
Stepping:               1
BogoMIPS:               6387.81
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtr

```

```

r pge mca cmov pat pse36 clflush mmx fxsr sse s
se2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtsc
p lm constant_tsc rep_good noopl xtopology tsc_r
eliable nonstop_tsc cpuid extd_apicid tsc_known
_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_
2 movbe popcnt aes xsave avx f16c rdrand hyperv
isor lahfm cmp_legacy svm cr8_legacy abm sse4
a misalignsse 3dnowprefetch osvw topoext perfct
r_core ssbd ibrs ibpb stibp vmmcall fsgsbase bm
il avx2 smep bmi2 erms invpcid rdseed adx smap
clflushopt clwb sha_ni xsaveopt xsavec xgetbv1
xsaves clzero xsaveerptr arat npt nrip_save tsc
_scale vmcb_clean flushbyasid decodeassists pau
sefilter pfthreshold v_vmsave_vmload umip vaes
vpclmulqdq rdpid fsrm

Virtualization features:
  Virtualization: AMD-V
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 128 KiB (4 instances)
  L1i: 128 KiB (4 instances)
  L2: 2 MiB (4 instances)
  L3: 8 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-7
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Not affected
  Retbleed: Not affected
  Spec rstack overflow: Vulnerable: Safe RET, no microcode
  Spec store bypass: Mitigation; Speculative Store Bypass disabled v
ia pretl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user
pointer sanitization
  Spectre v2: Mitigation; Retpolines; IBPB conditional; IBRS_
FW; STIBP always-on; RSB filling; PBRSE-eIBRS N
ot affected; BHI Not affected
  Srbds: Not affected
  Tsa: Vulnerable: No microcode
  Tsx async abort: Not affected

```

```

lenovo@Manyas-OSSP_Project$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.2 24232 15444 ?        Ss   13:51   0:01 /sbin/init
root         2  0.0  0.0   3180  2208 hvcd0    Sl+  13:51   0:00 /init
root         7  0.0  0.0   3180  2048 hvcd0    Sl+  13:51   0:00 plan9 --c
root        47  0.0  0.2 50424 16924 ?        S<s  13:51   0:00 /usr/lib/
systemd+   79  0.0  0.2 22416 14440 ?        Ss   13:51   0:00 /usr/lib/
root        85  0.3  0.2 35600 12544 ?        Ss   13:51   0:15 /usr/lib/
root       153  0.0  0.0 159348  1884 ?        Ssl  13:51   0:00 snapfuse
root       154  0.0  0.2 454144 13556 ?        Ssl  13:51   0:02 snapfuse
root       159  0.0  0.0 159348  1880 ?        Ssl  13:51   0:00 snapfuse
root       187  0.0  0.0   2888  2012 ?        Ss   13:51   0:00 /bin/sh /
root       188  0.0  0.0   4472  3116 ?        Ss   13:51   0:00 /usr/sbin
message+   189  0.0  0.0   8820  5376 ?        Ss   13:51   0:00 dbus-dae
root       193  0.0  0.5 44096 29688 ?        Ss   13:51   0:00 /usr/bin/
root       196  0.0  0.6 2220492 39504 ?        Ssl  13:51   0:03 /snap/sna
root       197  0.0  0.1 18440  9452 ?        Ss   13:51   0:00 /usr/lib/
syslog     214  0.0  0.0 220548  5644 ?        Ssl  13:51   0:00 /usr/sbin
root       230  0.0  0.0   5492  2796 ttyl    Ss+  13:51   0:00 /usr/sbin
_chrony     244  0.0  0.1 20512  7784 ?        S   13:51   0:00 /usr/sbin
root       252  0.0  0.5 123460 32648 ?        Ssl  13:51   0:00 /usr/bin/
_chrony     262  0.0  0.0 12096  2432 ?        S   13:51   0:00 /usr/sbin
root       393  0.0  0.0   8644  5596 ?        Ss   13:51   0:00 login --
akshith+   437  0.0  0.2 22772 13272 ?        Ss   13:51   0:00 /usr/lib/
akshith+   439  0.0  0.0 22920  4076 ?        S   13:51   0:00 (sd-pan)
akshith+   467  0.0  0.0   6268  5548 pts/1    Ss+  13:51   0:00 -bash
polkitd    1927  0.0  0.1 306444  8504 ?        Ssl  14:22   0:00 /usr/lib/
root      3942  0.0  0.0   3188  1120 ?        Ss   15:01   0:00 /init
root      3943  0.0  0.0   3204  1260 ?        S   15:01   0:00 /init
akshith+   3946  0.0  0.1  6784  6176 pts/0    Ss   15:01   0:00 -bash
root      4298  0.0  0.1 35604  6616 ?        S   15:05   0:00 (udev-mor
root      4299  0.0  0.1 35604  6616 ?        S   15:05   0:00 (udev-mor
akshith+   4305  0.0  0.0   7160  4412 pts/0    R+   15:05   0:00 ps aux
lenovo@Manyas-OSSP_Project$ echo $?
0

```

```

GNU nano 8.7.1
Linux is an operating system
Linux supports processes
Linux supports pipes
Linux supports signals
Linux supports system calls
Operating systems manage resources
Processes execute programs
Pipes connect commands
Signals notify processes

```

**Task 2:** To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation.

```

lenovo@Manya~/OSSP_Project$ mkdir documentation
lenovo@Manya~/OSSP_Project$ cd documentation
lenovo@Manya~/OSSP_Project/documentation$ nano architecture.md
lenovo@Manya~/OSSP_Project/documentation$ nano flow.txt
lenovo@Manya~/OSSP_Project/documentation$ cat flow.txt

```

```

graph TD
    START --> ReadUserCommand[Read User Command]
    ReadUserCommand --> ParseCommand[Parse the Command]
    ParseCommand --> IsPipe[Is there a Pipe?]
    IsPipe -- Yes --> CreatePipes[Create Pipes]
    IsPipe -- No --> ExecuteCommand[Execute Command]
    CreatePipes --> CreateProcesses[Create Processes]
    CreateProcesses --> ExecuteCommands[Execute Commands]
    ExecuteCommands --> WaitJobs[Wait for Jobs]
    WaitJobs --> DisplayOutput[Display Output]
    DisplayOutput --> END

```

```

lenovo@Manya~/OSSP_Project/documentation$ nano execution_path.md
lenovo@Manya~/OSSP_Project/documentation$ nano syscalls.md
lenovo@Manya~/OSSP_Project/documentation$ nano design_decisions.md
lenovo@Manya~/OSSP_Project/documentation$ ls
architecture.md  design_decisions.md  execution_path.md  flow.txt  syscalls.md
lenovo@Manya~/OSSP_Project/documentation$ |

```

```
GNU nano 8.7.1
# OSSP Project Architecture

## 1. User
The user enters commands through the terminal.

## 2. Shell
The shell receives and parses the commands.

## 3. Command Parser
The parser identifies commands, arguments, pipes and redirections.

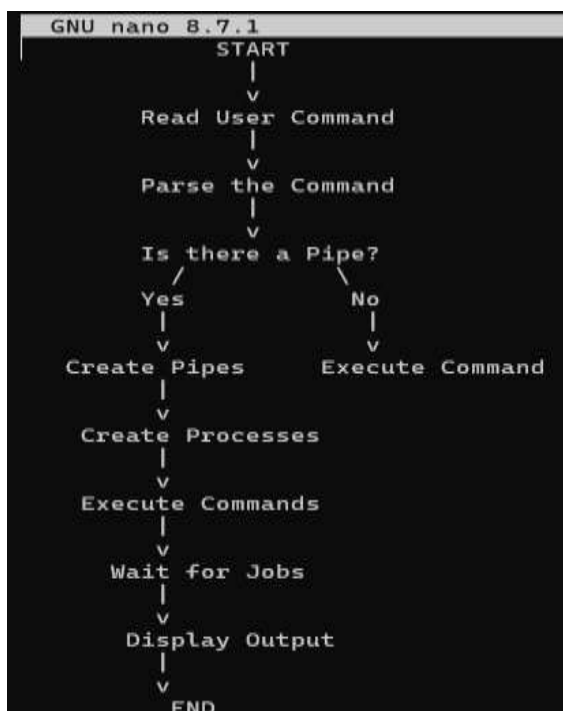
## 4. Process Creation
The shell creates child processes using system calls.

## 5. Execution
Child processes execute the required commands.

## 6. Communication
Pipes allow communication between processes.

## 7. Signals
Signals are used to control or notify processes.

## 8. Output
The final result is displayed on the terminal.
```



GNU nano 8.7.1

## # Execution Path

1. User enters a command.
2. Shell reads the command.
3. Shell parses the command.
4. If the command is external, a child process is created.
5. The child process executes the program.
6. Parent process waits for completion.
7. Output is displayed to the user.
8. The shell waits for the next command.

GNU nano 8.7.1

## # System Calls Used

### ## fork()

Creates a new child process.

### ## exec()

Replaces the current process with a new program.

### ## waitpid()

Allows the parent process to wait for a child process.

### ## pipe()

Creates a communication channel between processes.

### ## open()

Opens a file.

### ## read()

Reads data from a file.

### ## write()

Writes data to a file.

### ## close()

Closes an opened file descriptor.

### ## dup2()

Duplicates a file descriptor and is commonly used for redirection.

GNU nano 8.7.1

## # Design Decisions

1. Pipes are used for communication between processes.
2. fork() is used to create child processes.
3. exec() is used to execute external commands.
4. waitpid() is used to synchronize parent and child processes.
5. dup2() is used for input and output redirection.
6. Signals are used for process control.
7. Error messages are displayed when commands fail.

## Session-24

**Task 1:** To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

```
lenovo@Manya~$ cd ~/OSSP_Project
lenovo@Manya~/OSSP_Project$ mkdir -p src include tests docs
lenovo@Manya~/OSSP_Project$ ls
data.txt  docs  documentation  include  src  tests
lenovo@Manya~/OSSP_Project$ nano src/main.c
lenovo@Manya~/OSSP_Project$ nano build.sh
lenovo@Manya~/OSSP_Project$ chmod +x build.sh
lenovo@Manya~/OSSP_Project$ ./build.sh
Building OSSP Project...
Build successful.
lenovo@Manya~/OSSP_Project$ ./oss
OSSP Project Running Successfully
lenovo@Manya~/OSSP_Project$ nano README.md
lenovo@Manya~/OSSP_Project$ cat README.md
# OSSP Project

## Introduction
This project demonstrates operating system concepts using Linux.

## Features
- Process creation
- Process execution
- Pipes
- Signals
- Redirection
- Process synchronization

## Technologies
- C
- Linux
- GCC
- Bash

## Compilation

Run:

./build.sh

## Execution

Run:

./oss
```

```
lenovo@Manya~/OSSP_Project$ find . -maxdepth 2 -type f
./data.txt
./build.sh
./src/main.c
./ossdp
./documentation/flow.txt
./documentation/execution_path.md
./documentation/design_decisions.md
./documentation/syscalls.md
./documentation/architecture.md
./README.md
lenovo@Manya~/OSSP_Project$ ls documentation
architecture.md      execution_path.md  syscalls.md
design_decisions.md  flow.txt
```

```
GNU nano 8.7.1
#include <stdio.h>

int main()
{
    printf("OSSP Project Running Successfully\n");
    return 0;
}
```

```
GNU nano 8.7.1
#!/bin/bash

echo "Building OSSP Project..."

gcc src/main.c -o ossp

if [ $? -eq 0 ]
then
    echo "Build successful."
else
    echo "Build failed."
fi
```

```
GNU nano 8.7.1
# OSSP Project

## Introduction
This project demonstrates operating system concepts using Linux.

## Features
- Process creation
- Process execution
- Pipes
- Signals
- Redirection
- Process synchronization

## Technologies
- C
- Linux
- GCC
- Bash

## Compilation

Run:

./build.sh

## Execution

Run:

./ossdp
```

**Task2:** To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.

```
lenovo@Manya~/OSSP_Project$ mkdir -p demo
lenovo@Manya~/OSSP_Project$ nano demo/demo_commands.txt
lenovo@Manya~/OSSP_Project$ echo "Hello OSSP"
Hello OSSP
lenovo@Manya~/OSSP_Project$ sleep 10 &
[1] 5522
lenovo@Manya~/OSSP_Project$ jobs
[1]+  Running                  sleep 10 &
lenovo@Manya~/OSSP_Project$ echo "Linux Operating System" | grep Linux
Linux Operating System
[1]+  Done                      sleep 10
lenovo@Manya~/OSSP_Project$ echo "OSSP Project" > output.txt
lenovo@Manya~/OSSP_Project$ cat output.txt
OSSP Project
lenovo@Manya~/OSSP_Project$ echo "Pipeline Test" >> output.txt
lenovo@Manya~/OSSP_Project$ cat output.txt
OSSP Project
Pipeline Test
lenovo@Manya~/OSSP_Project$ ls
README.md  data.txt  docs          include  output.txt  tests
build.sh   demo      documentation  ossp     src
lenovo@Manya~/OSSP_Project$ find . -maxdepth 2 -type f
./data.txt
./build.sh
./demo/demo_commands.txt
./src/main.c
./oss
./documentation/flow.txt
./documentation/execution_path.md
./documentation/design_decisions.md
./documentation/syscalls.md
./documentation/architecture.md
./README.md
./output.txt
lenovo@Manya~/OSSP_Project$ ./build.sh
Building OSSP Project...
Build successful.
lenovo@Manya~/OSSP_Project$ ./oss
OSSP Project Running Successfully
lenovo@Manya~/OSSP_Project$ sleep 100 &
[1] 5730
lenovo@Manya~/OSSP_Project$ pgrep sleep
5730
lenovo@Manya~/OSSP_Project$ |
```

GNU nano 8.7.1

```
Demo 1: Basic command execution
Demo 2: Pipeline execution
Demo 3: Multiple processes
Demo 4: Input redirection
Demo 5: Output redirection
Demo 6: Signal handling
Demo 7: Error handling
```